MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

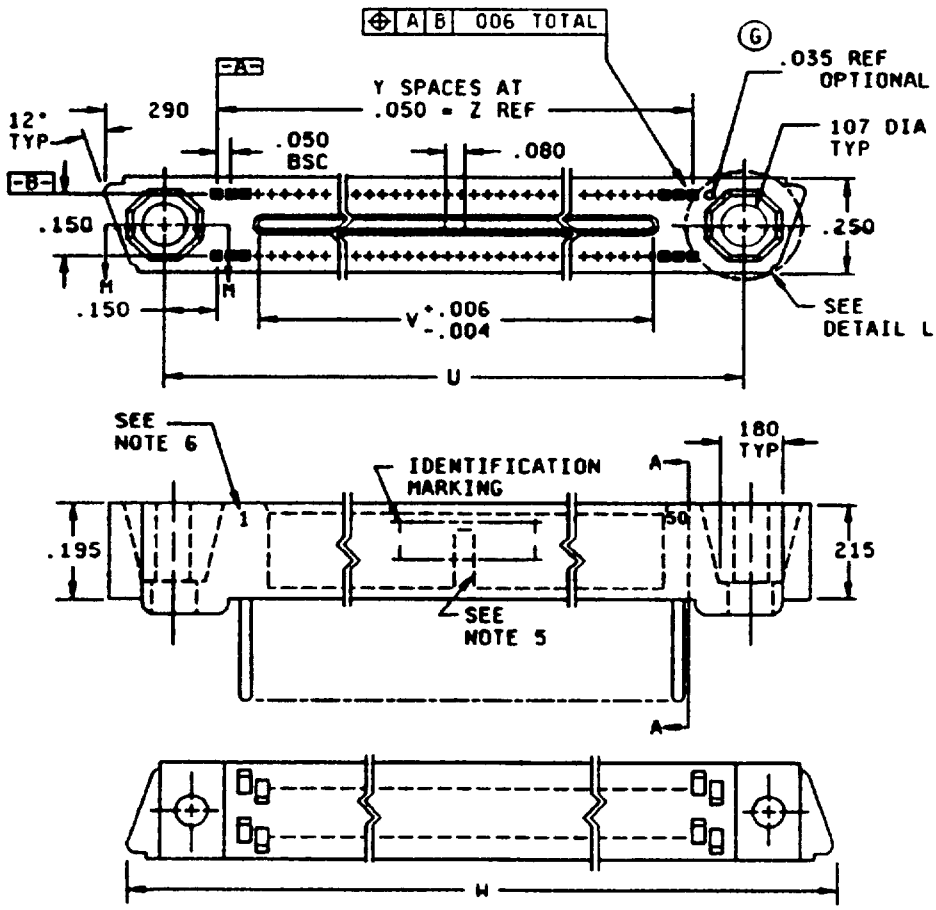
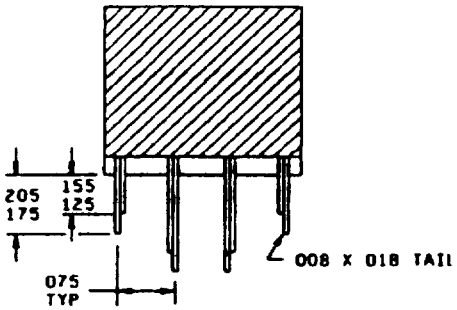


FIGURE 1. Connectors, receptacle, .050 (1.27 mm) spacing

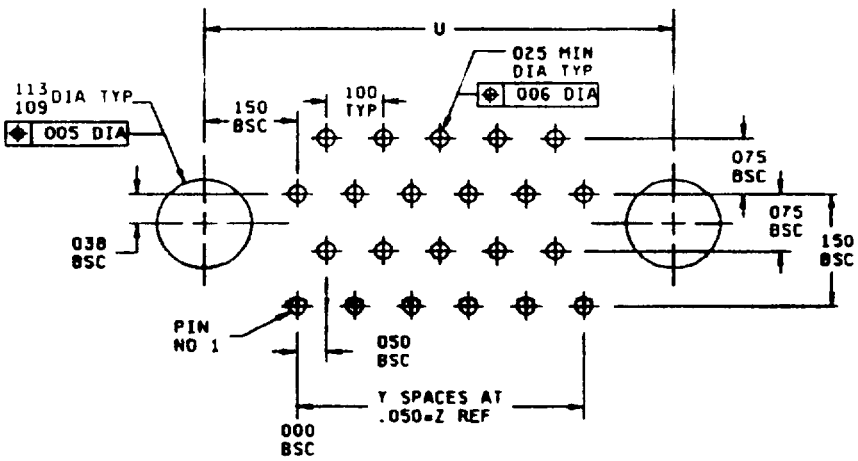
(G) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

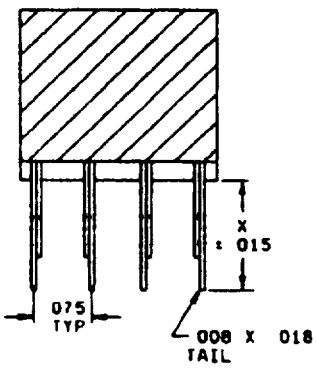
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SECTION A-A
TYPE I



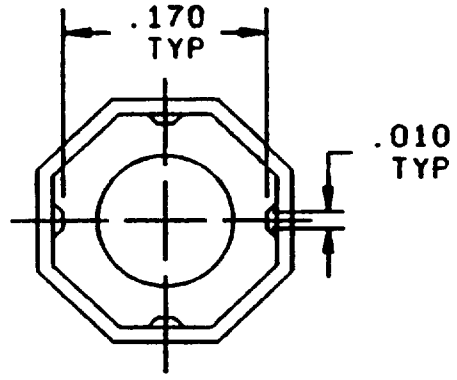
RECOMMENDED PRINTED WIRING BOARD HOLE
PATTERN, CONNECTOR SIDE



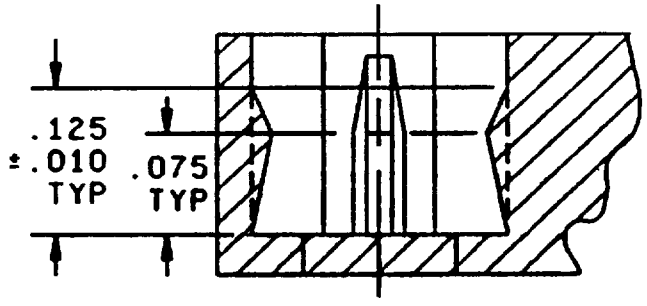
SECTION A-A
TYPE II

FIGURE 1. Connectors, receptacle .050 (1.27 mm) spacing - Continued.

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DETAIL L



SECTION M-M

Inches	mm	Inches	mm
.004	0.10	.109	2.77
.005	0.13	.113	2.87
.006	0.15	.125	3.18
.008	0.20	.150	3.81
015	0.38	.155	3 94
.018	0.46	.175	4 44
.025	0.64	.180	4.57
.038	0.97	.195	4.95
.050	1.27	.205	5.21
.075	1.98	.215	5.46
.080	2.03	.250	6.35
.098	2 49	.290	7.37
100	2.54		

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerance is ±.005 (0 13 mm) and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/117(USAF) and MIL-C-55302/118(USAF)
5 Barrier is not present on 20 through 50 position
6. Numbers indicating end positions stamped this side only.

FIGURE 1. Connectors, receptacle, .050 (1.27 mm) spacing - Continued

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TABLE I Dash numbers and dimensions

Type	Dash number	No of contacts	Dimensions 1/						Contact ident. numbers
			U	V	W	X	Y	Z	
I	01	20	.750 (19.05)	.300 (7.62)	1.040 (26.42)		9	450 (11.43)	1-10
I	02	30	1.000 (25 40)	.550 (13.97)	1.290 32.77		14	.700 (17 78)	1-15
I	03	40	1.250 (31.75)	800 (20 32)	1.540 (39.12)		19	.950 (24.13)	1-20
I	04	50	1 500 (38 10)	1 050 (26 67)	1 790 (45 47)		24	1.200 (30 48)	1-25
I	05	60	1.750 (44.45)	1 300 (33 02)	2 040 (51.82)		29	1 450 (36.83)	1-30
I	06	70	2 000 (50.80)	1 550 (39.37)	2 290 (58.17)		34	1 700 (43.18)	1-35
I	07	80	2.250 (57.15)	1 800 (45.72(	2.540 (64.52)		39	1.950 (49 53)	1-40
I	08	90	2.500 (63.50)	2.050 (52.07)	2.790 (70.87)		44	2.200 (55.88)	1-45
I	09	100	2.750 (69.85)	2 300 (58.42)	3.040 (77.22)		49	2.450 (62.23)	1-50
I	10	110	3.000 (76.20)	2.550 (64 77)	3.290 (83.57)		54	2.700 (68 58)	1-55
I	11	128	3.450	3.000 (76.20)	3.740 (95.00)		63	3.150 (80.01)	1-64
II	12	100	2.750 (69.85)	2.300 (58.42)	3 040 (77.22)	.300 (7.62)	49	2.450 (62.23)	1-50
II	13	110	3.000 (76.20)	2.550 (64.77)	3.290 (83.57)	.300 (7.62)	54	2.700 (68.58)	1-55
II	14	128	3.450 (87.63)	3.000 (76.20)	3.740 (95 00)	.190 (4.82)	63	3.150 (80.01)	1-64

1/ Metric equivalents are in parentheses.

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REQUIREMENTS

Design and construction

Dimensions and configurations: See figure 1.

(G) Material

Body: Insulator body shall be in accordance with MIL-C-55302 or MIL-M-24519 type GLCP-30F.

Contact material: Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768.

Plating: Contact plating shall be gold in accordance with MIL-C-55302 over nickel in accordance with MIL-C-55302 on the contact engagement area for .100 inch minimum length The termination area plating shall be tin lead over nickel in accordance with MIL-C-55302, to within 020 inch maximum of plastic body

Contact identification: See figure 1.

Oversized pin exclusion 0330 diameter pin

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by 30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact engagement and separation force:

Engagement force: Maximum force to engage maximum test pin 0190 +.0000, - 0001 diameter shall be 6 ounces.

Separation force: Minimum force to extract minimum test pin 0170 + 0000, -.0000, diameter shall be 10 ounces

Contact rating: 1.5 amperes per contact, maximum

Contact resistance: When mated with MIL-C-55302/117, the average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have a resistance exceeding .030 ohm When mated with MIL-C-55302/118, the average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: .75 pound, minimum.

Operating temperature: -65°C to +125°C.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only) (applicable to qualification and retention of qualification only)

Part number M55302/119- (dash number from table I)

Patent number 3,404,367 The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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CONCLUDING MATERIAL

Custodian:
Air Force - 17

Review activities
Air Force - 85, 99
DLA - ES

User activity:
Air Force - 19

Preparing activity
Air Force - 17

Agent
DLA - ES

(Project 5935-F405-03)

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